



EMPLOYEE ATTRITION PREDICTION

Developer's MVP Definition

Frontend & Backend Development Backlog, Responsibilities, Roadmap & Definition of Done

1. Developer Vision

As the Developer, my objective is to design, develop, integrate, and maintain both the frontend and backend components of the Employee Attrition Prediction platform. The system provides a responsive web interface for HR users and a secure, scalable backend that manages authentication, employee records, machine learning prediction, dashboard analytics, reporting, and seamless data exchange between the frontend and backend, enabling HR teams to make data-driven retention decisions.

Developer: Shiva Shankar

Delivery Window: 01 July 2026 – 15 July 2026 (5 Sprints)

Target Release: MVP Demo at the end of Sprint 5 (15-07-2026)

2. Target Users / Personas

Persona	Role	Developer Need	Key System Interaction
Priya	HR Business Partner	Securely access employee attrition predictions through an intuitive interface	Login, dashboard, employee risk prediction, reports
Rahul	HR Analyst	Manage employee records, perform bulk predictions, and generate reports	Employee CRUD, batch prediction, analytics dashboard, reports
Meera	HR Director	Monitor organizational attrition trends and analytics	Dashboard analytics, prediction history, reports, decision support

3. Problem Statement & Developer MVP Goal

The Problem

An employee attrition prediction system requires more than a machine learning model. HR users need an integrated web application that securely manages employee data, performs predictions, presents analytics through an interactive dashboard, and provides reports. The frontend and backend must work together seamlessly to deliver an end-to-end user experience.

Developer MVP Goal

Develop a complete full-stack application consisting of a responsive React frontend and a secure FastAPI backend. The application should support authentication, employee management, machine learning prediction, dashboard analytics, reporting, and frontend-backend integration, providing HR teams with an end-to-end employee attrition prediction workflow.

Success Metrics (Developer View)

- **UI/UX:** Responsive and user-friendly frontend is fully functional.
- **API:** REST APIs are implemented and documented using Swagger.
- **Security:** JWT authentication secures protected resources.
- **Data Management:** Employee CRUD operations work successfully.
- **AI Engine:** Machine learning APIs generate accurate predictions.
- **Insights:** Dashboard displays real-time analytics and visualizations.
- **Records:** Reports and prediction history are accessible.
- **Integration:** Frontend communicates successfully with backend APIs.
- **Workflow:** End-to-end workflow executes without manual intervention.

4. Developer Backlog (MoSCoW Prioritization)

Priority	Module	Developer User Story / Notes
Must Have	Frontend Project Setup	As a developer, I need a React project structure to build a responsive user interface.
Must Have	Backend Project Setup	As a developer, I need a modular FastAPI project for maintainable backend development.
Must Have	Database Integration	

Priority	Module	Developer User Story / Notes
		As a developer, I need SQLAlchemy models and database connectivity to manage application data.
Must Have	Authentication	As a developer, I need secure authentication so only authorized users can access the system.
Must Have	Employee Management	As an HR Analyst, I need to add, update, view, and delete employee records.
Must Have	Dataset Processing	As a developer, I need preprocessing APIs to ensure clean data for model training.
Must Have	Machine Learning Integration	As an HR user, I need accurate employee attrition predictions.
Must Have	Dashboard	As an HR Director, I need analytics dashboards with charts and statistics.
Must Have	Reports	As an HR Analyst, I need employee, prediction, and training reports.
Must Have	Frontend–Backend Integration	As a developer, I need the frontend to communicate seamlessly with backend APIs.
Should Have	Analytics Module	As HR Management, I need additional analytics for decision making.
Should Have	Logging	As a developer, I need application logs for monitoring and debugging.
Won't Have (Post-MVP)	HRMS Integration	Deferred for future releases.
Won't Have (Post-MVP)	Email Notification Service	Deferred after MVP validation.
Won't Have (Post-MVP)	Multi-Organization Support	Deferred until production release.
Won't Have (Post-MVP)	Advanced RBAC	Basic JWT authentication is sufficient for MVP.
	Mobile Application	Web application support only for MVP.

Priority	Module	Developer User Story / Notes
Won't Have (Post-MVP)		

5. Release Roadmap (Sprint View)

Sprint	Date	Developer Focus	Key Deliverable
Sprint 1	01-07-2026	Project setup, React & FastAPI initialization, database configuration, UI wireframes, Swagger documentation	Project foundation: Structure, database models, UI wireframes, API documentation
Sprint 2	04-07-2026	Authentication module, login interface, employee CRUD implementation	Core Access: Authentication module, Employee Management module
Sprint 3	07-07-2026	Machine learning integration, dataset processing, prediction APIs, prediction UI	AI Engine: Intelligent Prediction Module
Sprint 4	10-07-2026	Dashboard development, reports module, prediction history, backend modularization, frontend integration	Insights: Analytics Dashboard & Reporting Module
Sprint 5	13-07-2026 – 15-07-2026	System integration, frontend-backend testing, bug fixing, API validation, documentation	Final Delivery: Fully Integrated MVP Application ready for demonstration

6. Definition of Done (Developer MVP)

The Developer MVP is considered complete when:

- React frontend is fully functional.
- FastAPI backend is fully operational.
- REST APIs are implemented and documented using Swagger.
- JWT Authentication is functional.
- Employee CRUD operations work end-to-end.

- Dataset preprocessing and feature engineering are implemented.
- Machine learning model training and prediction execute successfully.
- Dashboard displays analytics and visualizations.
- Prediction history is stored successfully.
- Reports are generated correctly.
- Frontend communicates successfully with backend APIs.
- Backend routes are modularized into separate route files.
- End-to-end workflow (Login → Employee Management → Prediction → Dashboard → Reports) executes successfully.
- The complete system is demonstrated successfully during the Sprint 5 MVP presentation.

7. Key Risks & Developer Mitigation

Risk	Impact	Mitigation
Tight 5-sprint timeline	Limited development and testing time	Prioritize core MVP features and perform continuous sprint testing
Poor dataset quality	Reduced prediction accuracy	Apply preprocessing, validation, and feature engineering before training
Authentication vulnerabilities	Unauthorized access	Implement JWT authentication and secure protected APIs
Frontend–backend integration issues	UI unable to consume APIs	Standardize REST APIs, use Axios, and validate endpoints with Swagger
Increasing project complexity	Difficult maintenance	Adopt modular architecture for frontend components and backend routes/services
Machine learning performance	Slow or inaccurate predictions	Optimize preprocessing pipeline and evaluate multiple ML models

8. Stakeholders

- **Product Owner:** Akshay — Defines product vision, prioritizes backlog, and approves MVP scope.
- **Developer:** Shiva Shankar — Designs, develops, integrates, tests, and maintains the frontend and backend components of the application.

- **Machine Learning Team:** Develops and validates prediction models (Logistic Regression, Random Forest, XGBoost, SHAP).
- **End Users:** HR Business Partners, HR Analysts, HR Directors.
- **Project Evaluators:** Final Year Project – AI/ML Track Review Committee.